

UVA CS 4774: Machine Learning

S6: Lecture 30: Quiz Reviews

Dr. Yanjun Qi

University of Virginia
Department Of Computer Science



Q1

Q1: Given the definitions of A and B below, compute AB.



☒ Multiple choice

☒ $A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 \\ -1 & 0 \\ 5 & 2 \end{bmatrix}$

☐ Option 1

$$\begin{bmatrix} -5 & -1 \\ 17 & 8 \end{bmatrix}$$

☐ Option 2

$$\begin{bmatrix} -5 & 1 \\ -17 & 8 \end{bmatrix}$$

☐ Option 3

$$\begin{bmatrix} 9 & 1 \end{bmatrix}$$



Q2: For conformable matrices A and B, which of the following always holds?



☒ Multiple choice

- ☐ $(AB)^T = (B^T)(A^T)$
- ☐ $(AB)^T = AB$
- ☐ $(AB)^T = AB^T$
- ☐ Add option or [add "Other"](#)



Q3:



☒ Multiple choice

⋮

a matrix $D \in R^{5 \times 7}$, which of the following must always be true

- ☐ Rank(D) = 5
- ☐ Rank(D) = 7
- ☐ Rank(D) ≤ 5
- ☐ Rank(D) ≥ 5
- ☐ Rank(D) ≥ 7
- ☐ Add option or [add "Other"](#)



Answer key (2 points)



Required





Q2



1. In $f:X \rightarrow Y$, which represents the model?



☒ Multiple choice

☐ f



☐ X

☐ Y

☐ Add option or [add "Other"](#)



Answer key (2 points)



Required



2. How does unsupervised learning differ from supervised learning?



Multiple choice

- ☐ No input X is provided
- ☐ No label Y is provided
- ☐ Label y is continuous
- ☐ Label y is discrete
- ☐ Add option or [add "Other"](#)



Answer key (2 points)



Required



...

3. Generalization refers to how well your model performs on



☒ Multiple choice

- ☐ The testing set
- ☐ The training set
- ☐ Both the training and testing set
- ☐ Add option or [add "Other"](#)





4. The difference between classification and regression is?



☒ Multiple choice

- ☐ Different types of input
- ☐ Different types of output
- ☐ Different types of model
- ☐ Different types of programs
- ☐ Add option or [add "Other"](#)



Answer key (2 points)



Required





Q3



1. how can we find the best θ that minimizes loss function $J(\theta)$ in linear regression? (check all that apply)



Checkboxes

- ☐ Take derivative of $J(\theta)$ and set it to 0, solving for θ
- ☐ Calculate the gradient of $J(\theta)$ and use gradient descent iteratively
- ☐ Use binary search to find best θ
- ☐ Add option or [add "Other"](#)



Answer key (2 points)



Required



2. Suppose loss function $J(x) = x^2$, the initial value $x_0=1$, learning rate is 0.1, what will be x_1 if we apply gradient descent?



☒ Multiple choice

- ☐ $x_1 = -1$
- ☐ $x_1 = 0$
- ☐ $x_1 = 0.8$
- ☐ Add option or [add "Other"](#)



Answer key (2 points)



Required



...

3. Suppose we apply gradient descent with a learning rate that is too large. What is the most likely outcome?



☒ Multiple choice

- ☐ Convergence will be faster and guaranteed
- ☐ The algorithm may overshoot and fail to converge 
- ☐ The algorithm will converge to a local minimum instead of a global minimum
- ☐ The final solution will always be the closed-form solution
- ☐ Add option or [add "Other"](#)

 **Answer key** (2 points)



Required 



4. Comparing SGD, GD, minibatch-GD, what is the most important difference?



☒ Multiple choice

- ☐ Error metric/ Loss function
- ☐ The number of data used for each update
- ☐ Total number of epochs
- ☐ Add option or [add "Other"](#)

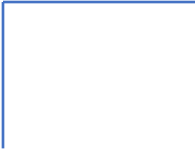


☒ Answer key (2 points)



Required





Q4

Q4 review

1. True or False? In a linear regression model, the number of parameters increase as the number of samples increase



Multiple choice

☐ True

☐ False

☐ Add option or [add "Other"](#)



Answer key (2 points)



Required



Q4 review



2. Which data split should the model never be trained on? ()



Multiple choice

☐ Training Set

☐ K fold Split

☒ Test Set



☐ Add option or [add "Other"](#)



Answer key (2 points)



Required



Q4 review

...

3. Which of the following is an advantage of k-fold cross validation over the simple validation set method for model selection?



Multiple choice

- ☐ Requires fewer computations
- ☐ Provides a lower-variance estimate of model performance
- ☐ Uses less training data overall
- ☐ Always achieves lower test error
- ☐ Add option or [add "Other"](#)



Answer key (2 points)



Required



Q4 review



4. Suppose a model has low training error but very high test error. Which of the following strategies could help? (check all that apply)



Checkboxes

☐ Use a more complex learner

☐ Add more training data



☐ Reduce the number of features



☐ Add option or [add "Other"](#)



Answer key (2 points)



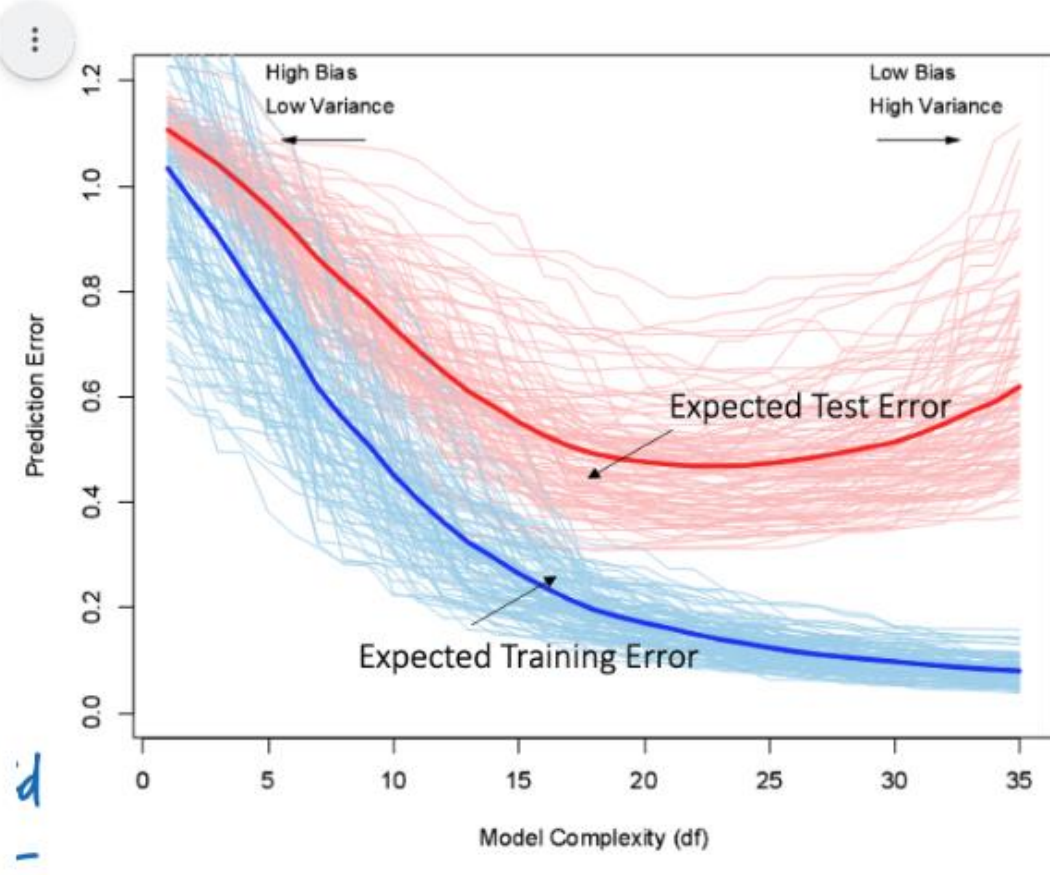
Required



5. Consider the below diagram. If you have a model with complexity of 5 degrees of freedom ($d=5$), it will face:



☒ Multiple choice



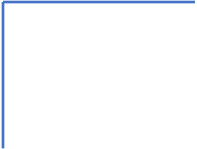
☐ Underfitting

☐ Overfitting

☐ Neither

☐ Add option or add "Other"





Q4

Q5 Review

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1. Which of the following best explains why Lasso Regression can perform feature selection?



Multiple choice

- ☐ It penalizes squared coefficients
- ☐ It can set some coefficients exactly to zero
- ☐ It increases the training error
- ☐ It reduces both training and test errors simultaneously
- ☐ Add option or [add "Other"](#)



Q5

2. Suppose we scale the weight feature from kilograms to grams in a KNN classifier. What effect does this have if we don't normalize the data?



☒ Multiple choice

- ☐ No effect, because KNN is scale-invariant
- ☐ Distance measure may get dominated by the weight feature
- ☐ The algorithm will automatically adjust feature ranges
- ☐ Model complexity will increase
- ☐ Add option or [add "Other"](#)



Q5



3. Which of the following is a disadvantage of increasing k in KNN when used for classification?



☒ Multiple choice

- ☐ Model becomes more sensitive to noise
- ☐ Model predictions become less stable
- ☐ Model introduces more smooth decision boundary
- ☐ The decision boundary becomes more irregular
- ☐ Add option or [add "Other"](#)



Q5

...

4. In a vanilla (i.e. regular) k-nearest neighbor classifiers, as k increases, the model



☒ Multiple choice

- ☐ becomes more complex and may overfit
- ☐ becomes more simple and may underfit
- ☐ Add option or [add "Other"](#)



Q5

...

5. Select method that prevent overfitting



☒ Checkboxes

☐ Keep increasing model complexity

☐ Keep adding training epochs

☐ Use proper regularizer

☐ Add option or [add "Other"](#)



Q5

6. (Extra Credit) In k-nearest neighbor classifiers, suppose k is fixed, as the number of training samples (N) increase, the time it takes to test a new sample:



☒ Multiple choice

- ☐ Increases
- ☐ Decreases
- ☐ Stays the same
- ☐ Add option or [add "Other"](#)



7. (Extra Credit)

Q5



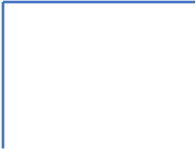
Checkboxes

Assume we have n training (x, y) pairs, and each x has 5 features. After training with ridge regression loss shown as follows, we get $w_1 = 0.8$, $w_2 = -1$, $w_3 = 0.01$, $w_4 = 0.001$, $w_5 = 0.1$, choose two most important features.

⋮

$$L(W) = \frac{1}{n} \sum_{i=1}^n [y_i - (w_1 x_{i,1} + w_2 x_{i,2} + w_3 x_{i,3} + w_4 x_{i,4} + w_5 x_{i,5} + b)]^2 + \beta ||W||_2^2$$

☐ x1☐ x2☐ x3☐ x4☐ x5☐ Add option or [add "Other"](#)



Q6

1. In L2-regularized linear regression, increasing the regularization strength λ generally...



Multiple choice

- ☐ decreases bias and increases variance
- ☐ increases bias and decreases variance
- ☐ decreases both bias and variance
- ☐ increases both bias and variance
- ☐ Add option or [add "Other"](#)



Answer key (2 points)



Required

2. For k-NN on a fixed dataset, which k is most vulnerable to overfitting and what happens to the training error?



☒ Multiple choice

- ☐ k=1; training error ≈ 0
- ☐ k=15; training error ≈ 0
- ☐ k=15; training error very high
- ☐ k=1; training error very high
- ☐ Add option or [add "Other"](#)



☒ **Answer key** (2 points)



Required ☒

3. Which actions are appropriate when a model shows high variance? (select all that apply)



☒ Checkboxes

- ☐ Use a simpler model ✓
- ☐ Add regularization ✓
- ☐ Get more training data ✓
- ☐ Reduce feature set ✓
- ☐ Build an ensemble (bagging) ✓
- ☐ Add option or [add "Other"](#)

☒ Answer key (2 points)



Required ☐

4. In the diagram below, the red center shows the true values, and the blue points show the predictions. The model has:



☒ Multiple choice

B *I* U



- ☐ High bias, low variance
- ☐ High bias, high variance
- ☐ Low bias, low variance
- ☐ Low bias, high variance
- ☐ Add option or [add "Other"](#)





5. In a vanilla (i.e. regular) k-nearest neighbor classifiers, as k increases, the model



☒ Multiple choice

- ☐ becomes more complex and may overfit
- ☐ becomes more simple and may underfit
- ☐ Add option or [add "Other"](#)



☒ Answer key (2 points)



Required ☒

...

6. Select two methods that prevent overfitting



Checkboxes

- ☐ Increase number of training data
- ☐ Keep increasing model complexity
- ☐ Keep adding training epochs
- ☐ Use proper regularizer
- ☐ Add option or [add "Other"](#)



Answer key (2 points)



Required





7. If A and B are independent random variables, then $P(A \text{ and } B) =$



☒ Multiple choice

☐ $P(A) * P(B)$



☐ $P(A) + P(B)$

☐ Add option or [add "Other"](#)

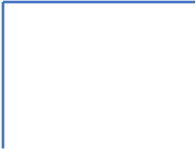


Answer key (2 points)



Required





Q7



1. Logistic regression is considered a discriminative model because it:



Multiple choice

- ☐ Models the joint probability $P(X,Y)$ directly.
- ☐ Models the conditional probability $P(Y|X)$ directly. 
- ☐ Assumes features are independent given the class label.
- ☐ Uses Bayes' theorem to estimate priors and likelihoods.
- ☐ Add option or [add "Other"](#)



Answer key (2 points)



Required





2. At the decision boundary of logistic regression, $P(y = 1 | x) = P(y = 0 | x) = ?$



Multiple choice

☐ 1

☐ -1

☐ 0.5

☐ 0

☐ Add option or [add "Other"](#)



Answer key (2 points)



Required





3. True or False? Logistic regression models have a linear decision boundary.



Multiple choice



☐ True

☐ False

☐ Add option or [add "Other"](#)



Answer key (2 points)



Required



4: Logistic regression is:



Multiple choice

☐ An unsupervised algorithm

☐ A supervised algorithm



☐ Add option or [add "Other"](#)



Answer key (2 points)



Required





5. The decision boundary of logistic regression is:



Multiple choice



- ☐ Always linear in feature space
- ☐ Always nonlinear in feature space
- ☐ Quadratic in feature space
- ☐ Randomly determined by initialization
- ☐ Add option or [add "Other"](#)



Answer key (2 points)



Required



6. In a neural network, the activation function introduces:



Multiple choice

- ☐ Linearity
- ☐ Non-linearity
- ☐ Bias
- ☐ Regularization
- ☐ Add option or [add "Other"](#)



Answer key (2 points)



Required



...

8. True or False? Summation of the outputs of a softmax function is 1.



Multiple choice

☐ True



☐ False

☐ Add option or [add "Other"](#)

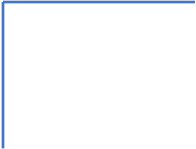


Answer key (2 points)

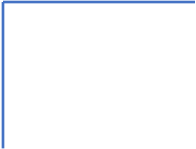


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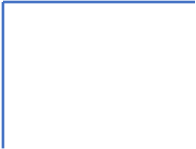




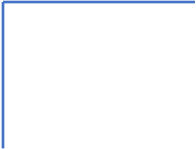
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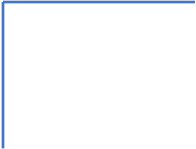
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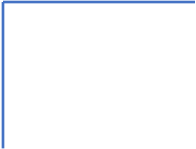
Q10



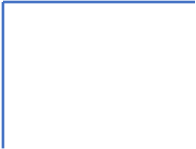
Q11



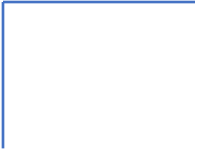
Q12



Q13



Q14



Q15

